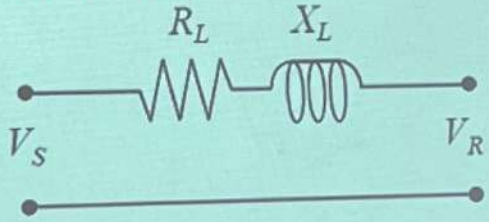
- **Transmission lines are represented by an equivalent circuit with parameters on a per-phase basis**
  - Voltages are expressed as phase-to-neutral
  - Currents are expressed for one phase
  - The three phase system is reduced to an equivalent single-phase
- **All lines are made up of distributed series inductance and resistance, and shunt capacitance and conductance**
  - Line parameters: R, L, C, & G
- **Three types of models**
  - depend on the length and the voltage level
  - short, medium, and long length line models

Short Transmission Line Model

- **The short transmission line model may be used when**
  - The line length is less than 50 miles (80 km), or
  - The line voltage is not over 69 kV
- **Modeling of the transmission line parameters**
  - The shunt capacitance and conductance are ignored
  - The line resistance and reactance are treated as lumped parameters
- **Circuit of the short model**



Two-Port Representation

- **Circuit Equations:**

$$V_S = V_R + Z_{line} I_R$$

$$I_S = I_R$$

  - Matrix representation:

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} 1 & Z_{line} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

  - ABCD values:

$$A = 1$$

$$B = Z_{line}$$

$$C = 0$$

$$D = A = 1$$

Means and requirements on an electrical supply system

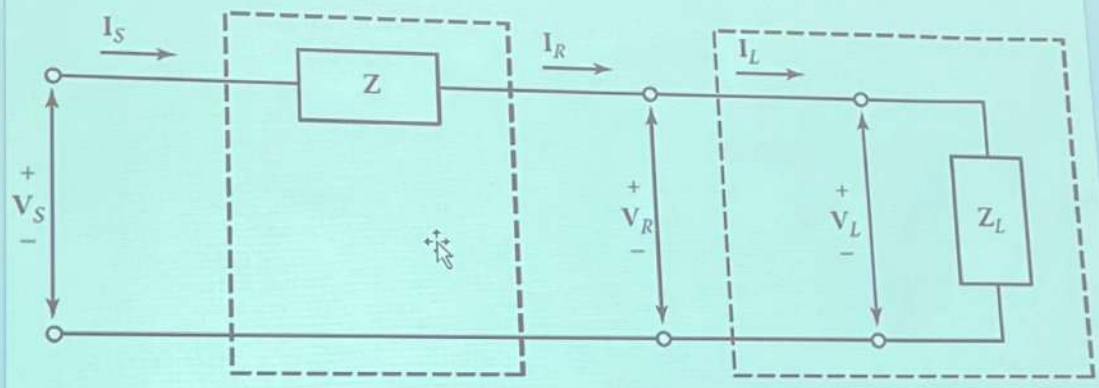
May include following features:

- Specified voltage levels
- Limited harmonic content (quality of supply)
- Method of supply
- Safety of supply
- Reliability of supply
- Maintenance
- Electrical protection
- Back-up supply (UPS)
- DC supply

a short transmission line that is terminated by a load of 200 kVA at a lagging power factor of 0.866 at 2.4 kV If the line impedance is $2.07 + j0.661\ \Omega$, calculate

(a) Sending-end current
(b) Sending-end voltage
(c) Input impedance
(d) Real and reactive power loss in the line

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$



$$Z = 2.07 + j0.661 = 2.173\angle 17.7^\circ\ \Omega$$

$$\mathbf{I}_R = \mathbf{I}_S = \mathbf{I}_L \qquad \mathbf{V}_R = \mathbf{Z}_L \mathbf{I}_R$$

$$\mathbf{S}_R = 200\angle 30^\circ = 173.2 + j100\ \text{kVA} \qquad \mathbf{V}_L = 2.4\angle 0^\circ\ \text{kV}$$

$$\mathbf{I}_L^* = \frac{\mathbf{S}_R}{\mathbf{V}_I} = \frac{200\angle 30^\circ}{2.4\angle 0^\circ} = 83.33\angle 0^\circ \text{ A}$$

$$\mathbf{I}_L = 83.33\angle -30^\circ \text{ A}$$

$$\mathbf{I}_S = \mathbf{I}_R = \mathbf{I}_L = 83.33\angle -30^\circ \text{ A}$$

$$\mathbf{Z}_L = \frac{\mathbf{V}_L}{\mathbf{I}_L} = \frac{2.4\times 10^3\angle 0^\circ}{83.33\angle -30^\circ} = 28.8\angle 30^\circ\ \Omega$$

$$\mathbf{V}_R = \mathbf{Z}_L\mathbf{I}_R = 28.8\angle 30^\circ \times 83.33\angle -30^\circ = 2404\angle 0^\circ \text{ kV}$$

$$\mathbf{V}_S = \mathbf{A}\mathbf{V}_R + \mathbf{B}\mathbf{I}_R$$

$$= 2400\angle 0° + 2.173\angle 17.7° \times 83.33\angle -30°$$

$$= 2576.9 - j38.58$$

$$= 2577.2\angle -0.9° \text{ V}$$

(c) The input impedance is

$$\mathbf{Z}_{in} = \frac{\mathbf{V}_S}{\mathbf{I}_S} = \frac{\mathbf{A}\mathbf{V}_R + \mathbf{B}\mathbf{I}_R}{\mathbf{C}\mathbf{V}_R + \mathbf{D}\mathbf{I}_R}$$

$$= \frac{2577.2\angle -0.9°}{83.33\angle -30°} = 30.93\angle 29.1° \ \Omega$$

(d) The real and reactive power loss in the line:

$$\mathbf{S}_L = \mathbf{S}_S - \mathbf{S}_R$$

$$\mathbf{S}_S = \mathbf{V}_S \mathbf{I}_S^* = 2.577.2\angle -0.9° \times 83.33\angle 30°$$

$$= 214{,}758\angle 29.1° \text{ VA}$$

$$\mathbf{S}_L = 214{,}758\angle 29.1° - 200{,}000\angle 30°$$

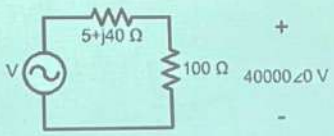
$$= 14{,}444.5 + j4{,}444.4 \text{ VA}$$

that is, the active power loss is 14,444.5 W, and the reactive power loss is 4444.4 vars.

Q1) What should be done to provide reliable electric power to customers?

Answer:
To provide electric power reliably to consumers, it becomes necessary to manage the generation, transmission, and consumption of electric power efficiently. Since the load demand is controlled by the consumers and may change randomly depending upon consumers' wishes, and since the generation must match this demand (and losses) instant by instant, an elaborate system of controls is put in place at generators and Energy Management Centers on the power grid.

Q2.) Solve for the total power delivered by the source, calculate the power factor, power factor angle real and reactive power, and V



$$I = \frac{40000\angle 0^\circ}{100\angle 0^\circ} = 400\angle 0^\circ \text{ Amps}$$

$$S = VI^* = 44.9\angle 20.8^\circ \cdot 400\angle 0^\circ$$

$$S = 17.98\angle 20.8^\circ \text{ MVA} = 16.8 + j6.4 \text{ MVA}$$

$$V = 40000\angle 0^\circ + (5 + j40) \cdot 400\angle 0^\circ$$

$$V = 42000 + j16000 = 44.9\angle 20.8^\circ \text{ kV}$$

88. Which of the following will need the highest level of illumination ?

a. Proof reading
b. Bed rooms
c. Hospital wards
d. Railway platforms

89. Which of the following will need lowest level of illumination ?

a. Displays
b. Fine engraving
c. Railway platform
d. Auditoriums

103. What are the types of over voltages?

Internal and External Overvoltages

1 .What is the use of ground wire?

It shields the transmission line conductor from induced charges, from clouds as well as from a lightning discharge.

2. Incoming _____ should have a _____ switch and a _____ _____. A disconnect switch is used to provide isolation since it cannot _____ load current. A circuit breaker is used as a _____ device to interrupt _____ currents automatically. Both _____ and _____ _____ may be operated locally or _____ from a supervisory control center.

3. What can trigger Blackouts in a Power Distribution System?

Faults , Equipment damage , Human errors , Natural disasters , Sabotage

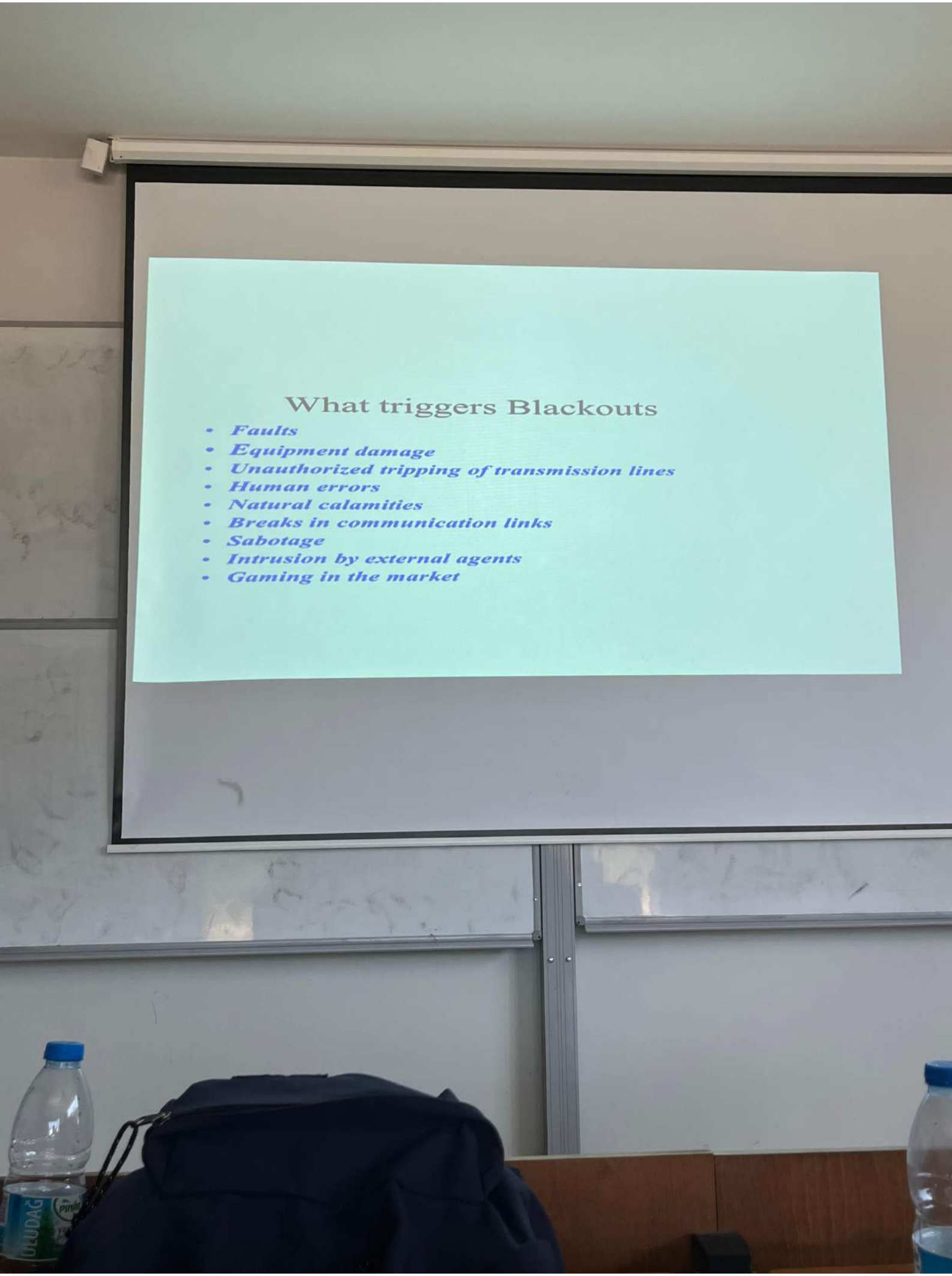
Gaming in the Market , Intrusion by external agents , Unauthorized tripping of TLs

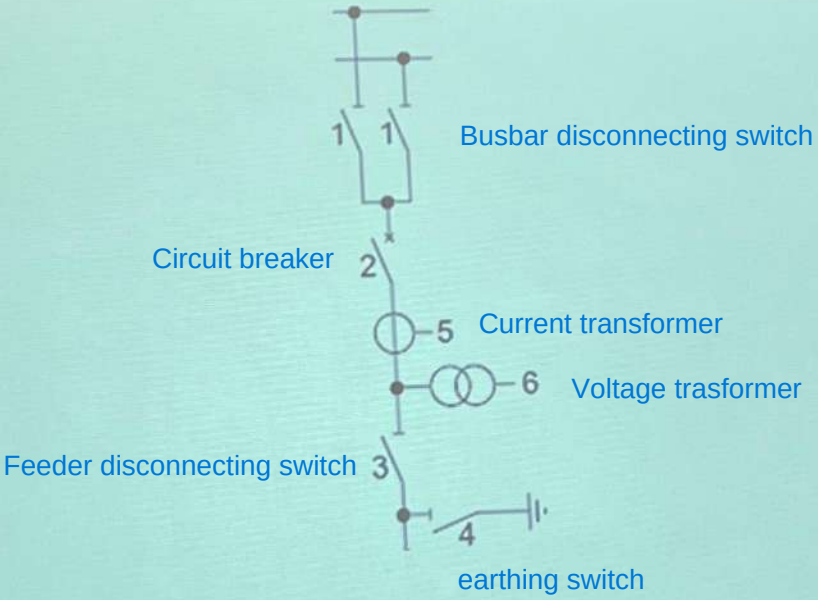
Necessity of Earthing:

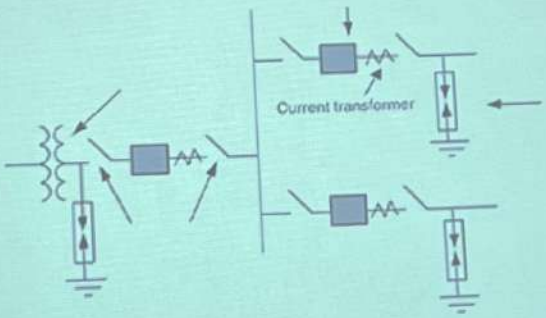
- To protect the operating personnel from danger of shock in case they come in contact with the charged frame due to defective insulation.
- To maintain the line voltage constant under unbalanced load condition.
- Protection of the equipments
- Protection of large buildings and all machines fed from overhead lines against lightning.

Methods of Earthing:

- Plate earthing
- Pipe earthing
- Strip or wire earthing
- Rod earthing


What triggers Blackouts
• Faults
• Equipment damage
• Unauthorized tripping of transmission lines
• Human errors
• Natural calamities
• Breaks in communication links
• Sabotage
• Intrusion by external agents
• Gaming in the market


1
1
Busbar disconnecting switch
Circuit breaker
2
5
Current transformer
6
Voltage trasformer
Feeder disconnecting switch
3
4
earthing switch


Current transformer

A good system whether overhead or underground should fulfill the following requirements

The voltage at the consumer's premises must be maintained within ± 4 or ± 6% of the declared voltage, the actual value depending on the type of load*.

The loss of power in the system itself should be a small percentage (about 10%) of the power transmitted.

The maximum current passing through the conductor should be limited to such a value as not to overheat the conductor or damage its insulation.

The insulation resistance of the whole system should be very high so that there is no undue leakage or danger to human life.

Different methods of feeding a distributor are given below :

feeding at one end

feeding at both ends with equal voltages

feeding at both ends with unequal voltages

feeding at some intermediate point

By which of the following systems electric power may be transmitted?

(a) Overhead system

(b) Underground system

(c) Both (a) and (b)

(d) None of the above

1. a) Fill in the names of the auxiliary equipment in the connection diagram.

b) Fill in the blanks. The station is protected from lightning and switching surges by a _____.
_____ ( ) is a large switch that interrupts the load and fault current. Fault detection systems automatically open the _____, but it can be operated manually. _____ provides visible circuit separation and permits _____ maintenance. It can be operated only when the _____ is open, in no-load condition.

Incoming line should have a disconnecting switch and a circuit breaker. A disconnect switch is used to provide isolation since it cannot interrupt load current. A circuit breaker is used as a protective device to interrupt faulty currents automatically. Both disconnect and circuit breaker may be operated locally or remotely from a supervisory control center. switch

Low-voltage cables are meant for use up to

(a)1.1 kV

(b)3.3 kV

(c)6.6 kV

(d) 11 kV

Supplying reactive power locally leads to decreased line current, which results in

- Decrease line loss
- Ability to use smaller wires
- Less voltage drop across the line

Advantages of 3ϕ Power

- Can transmit more power for same amount of wire (twice as much as single phase)
- Torque produced by 3ϕ machines is constrant
- Three phase machines use less material for same power rating
- Three phase machines start more easily than single phase machines